

1. BASIC NOTIONS ON SUBSTITUTIONS

1.1. Sequences, words, languages.

- $\mathcal{A}$ is a finite set that shall be called an *alphabet*.
- A *word* or *block* is a finite string of elements in $\mathcal{A}$.
- The set of all (finite) words over $\mathcal{A}$ is denoted by $\mathcal{A}^*$.
- The empty word is ϵ .
- If $W = w_1 \dots w_s$ then s is the *length* of W and is denoted by $|W|$.
 $|W|_a$ is the number of occurrences of $a \in \mathcal{A}$ appearing in W .
- A (*one-sided*) *sequence* of elements of $\mathcal{A}$, also called a (*right*) *infinite word* on $\mathcal{A}$, is an element $u = (u_n)_{n \geq 0} \in \mathcal{A}^{\mathbb{N}}$.
A *two-sided sequence* in $\mathcal{A}$, also called a *biinfinite word*, is an element $u = (u_n)_{n \in \mathbb{Z}} \in \mathcal{A}^{\mathbb{Z}}$.
- A word $v = v_1 \dots v_r$ is said to *occur* at *position* (*index* or *rank*) m in a sequence or finite word $u = (u_n)$ if

$$u_m = v_1, \dots, u_{m+r-1} = v_r.$$

We say v is a *factor* of the u .

- The *language* (respectively *language of length n*) of the sequence u , denoted by $\mathcal{L}(u)$ (respectively $\mathcal{L}_n(u)$), is the set of all words (respectively of length n) which occur in u .
- A sequence u is *recurrent* if every factor occurs infinitely often.
A sequence u is *periodic* (respectively *ultimately periodic*) if there exists a positive integer T such that

$$\forall n, u_n = u_{n+T}, \quad \text{respectively } \exists n_0 \forall |n| \geq n_0, u_n = u_{n+T}.$$

In both cases we say u is *shift-periodic*.

1.2. Complexity function.

- The *complexity function* p_u of a sequence u from a finite alphabet is

$$p_u(n) = \text{Card } \mathcal{L}_n(u), \quad n \in \mathbb{N},$$

i.e. the number of different words of length n in u .

Then $1 \leq p_u(n) \leq d^n$ where the alphabet has cardinality d .

- **Theorem.** *If u is ultimately periodic then p_u is bounded.*
If there exists an integer n such that $p_u(n) \leq n$ (in particular if p_u is bounded), then u is ultimately periodic.
- The *topological entropy* of u is the exponential growth rate of the complexity of u as the length increases:

$$H_{\text{top}}(u) = \lim_{n \rightarrow \infty} \frac{\log_d p_u(n)}{n},$$

where d is the cardinality of the alphabet $\mathcal{A}$.

The existence of the limit follows from the subadditivity of the function $n \mapsto \log_d p_u(n)$.

- The topological entropy of u equals the topological entropy of its associated dynamical system. See below.

1.3. Symbolic dynamical systems.

- $\mathcal{A}^{\mathbb{N}}$ has the product topology from the discrete topology on each copy of $\mathcal{A}$. It is defined by the metric

$$u \neq v \implies d(u, v) = 2^{-\min\{n \in \mathbb{N} : u_n \neq v_n\}}.$$

Similarly for $\mathcal{A}^{\mathbb{Z}}$

- The (left) shift operator is denoted by S .
- The orbit of u is $\mathcal{O}(u)$.
The closure is $X_u := \overline{\mathcal{O}(u)}$. It is compact and S is continuous on X_u . It is finite iff u is shift-periodic.
 $(\overline{\mathcal{O}}, S)$ is called the *symbolic dynamical system* associated with u .
- For $u, w \in \mathcal{A}^{\mathbb{N}}$ the following are equivalent
 - $w \in X_u$
 - $\exists (k_n)_{n \geq 0}$ s.t. $w_0 \dots w_n = u_{k_n} \dots u_{k_n+n} \forall n$.
 - $\mathcal{L}(w) \subset \mathcal{L}(u)$.
- u is recurrent iff u is a limit point of X_u iff S is onto on X_u .
- The *cylinder set* $[w_0 \dots w_r]$ is the set of extensions of $[w_0 \dots w_r]$.
Cylinder sets are clopen. Clopen sets are finite unions of cylinders.
- Extend $\mathcal{A}$ to $\mathcal{B}$ by adding a new letter. Then $\mathcal{A}^* \cup \mathcal{A}^{\mathbb{N}}$ is a closed subset of $\mathcal{B}^{\mathbb{N}}$ in a natural way.
- (*minimality* — same as minimality in a topological dynamical system— return to this in Section 5)

1.4. Topological dynamical systems. (p17)

- A *topological dynamical system* (X, T) is a compact metric space X with a continuous onto map $T : X \rightarrow X$. (Sometimes the “onto” is dropped.)
- (X, T) is *minimal* if for all $x \in X$ the orbit $\mathcal{O}$ is dense in X .
- (X, T) is minimal
 - iff X contains no nontrivial closed subset E s.t. $T(E) \subset E$.
 - iff for every nonempty open U , $\bigcup_{n \geq 0} T^{-n}(U) = X$.
- $f : T \rightarrow T$ is *T-invariant* if $\forall x \in X$, $f(Tx) = f(x)$.
If f is continuous and (X, T) is minimal then f is T -invariant iff f is constant.
- The set of Borel probability measures on X is a convex subset of the dual space of $\mathcal{C}(X)$, is compact and metrizable in the weak-star topology.
- A Borel measure μ on X is invariant if $T\mu = \mu$.
 (X, T) has an invariant probability measure. (Take any $x \in X$ and any limit point of $\frac{1}{N} \sum_{n < N} \delta_{T^n x}$.)
- (X, T) is *uniquely ergodic* if there exists exactly one T -invariant Borel prob measure over X .
- (X, S) and (Y, T) are *topologically equivalent* (or *topologically isomorphic*) if $\exists$ a homeomorphism $f : X \rightarrow Y$ which *conjugates* S and T , i.e. $f \circ S = T \circ f$.
- Example: If G is a compact group and T is mult by $a \in G$ then (X, T) is a topological dynamical system.
The Haar measure over G is invariant.

(X, T) is minimal iff $\{a^n : n \in \mathbb{N}\}$ is dense.

- Example: In particular, toral translations of $\mathbb{T}^d = \mathbb{R}^d / \mathbb{Z}^d$ by $(\alpha_1, \dots, \alpha_d)$ are topological dynamical systems.

They are minimal iff $1, \alpha_1, \dots, \alpha_d$ are rationally independent.

1.5. Substitutions. (p7)

- A substitution is a map $\sigma : \mathcal{A} \rightarrow \mathcal{A}^* - \{\epsilon\}$.
It extends to a map $\sigma : \mathcal{A}^* \rightarrow \mathcal{A}^*$ (with $\sigma(\epsilon) = \epsilon$), and to maps over $\mathcal{A}^{\mathbb{N}}$ and $\mathcal{A}^{\mathbb{Z}}$.
- The Fibonacci substitution is $\sigma(a) = ab, \sigma(b) = a$.
(See also web notes by Wright).

1.6. Abelianization. (p8)

- The *Incidence Matrix* associated to the alphabet $\mathcal{A}$ (with d symbols) and the substitution σ is the $d \times d$ matrix M_σ for which the i th entry in the j th column is the number of times a_i occur in $\sigma(a_j)$.
That is, $M_\sigma(i, j) = |\sigma(a_j)|_{a_i}$.
- The i th entry in the j th column of $(M_\sigma)^n$ is the number of times a_i occurs in $\sigma^n(a_j)$.
- The Fibonacci substitution has $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$

1.7. Measure-theoretic dynamical systems. (p19)

- Defn: $(X, T, \mu, \mathcal{B})$ is a *measure-theoretic dynamical system* if μ is a prob measure on a σ -algebra $\mathcal{B}$ of subsets of X and $T : X \rightarrow X$ is a Borel measurable map preserving μ .
- Suppose (X, T) is a topological dynamical system, μ is a T -invariant Borel prob measure, $\mathcal{B}$ is the set of Borel sets. Then $(X, T, \mu, \mathcal{B})$ is a measure-theoretic dynamical system.
[?? The article seems to require (X, T) be uniquely ergodic]
- $(X, T, \mu, \mathcal{B})$ is *ergodic* if every T -invariant Borel set has full or zero measure.
[?? This seems to require that X be a topological space]
- Ergodicity is equivalent to each of the following:
 - every Borel B such that $\mu(T^{-1}B \Delta B) = 0$ has zero or full measure;
 - every Borel B such that $\mu(B) > 0$ satisfies $\mu(\bigcup_{n \geq 0} T^{-n}B) = 1$.
- Suppose (X, T) is a uniquely ergodic topological system with corresponding measure μ . Then (X, T, μ) is ergodic.
- Suppose $(X, T, \mu, \mathcal{B})$ is a measure-theoretic dynamical system. Then T is ergodic iff every T -invariant measurable function is constant a.e.
- Let G be a compact metric group. Let T be rotation by $a \in G$. Then the following are equivalent:
 - T is minimal;
 - T is ergodic;
 - T is uniquely ergodic;
 - $\{a^n : n \geq 0\}$ is dense in G .

In particular, from the last, G is abelian.